ClassHub Smart Classroom Management System

Eliminating WhatsApp Chaos in Academic Communication

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Live Demo: http://16.171.176.40:5001/

Problem Statement

- Unorganized communication via WhatsApp
- Important updates get lost
- No proper system for: Attendance, Assignments, Timetable, Internal marks
- Lack of structured student-faculty interaction

Result: Chaotic, inefficient academic communication

Solution – ClassHub

A centralized platform providing:

- Organized announcements
- Assignment management
- Live timetable
- Attendance tracking
- Internal marks management
- Performance analytics

One platform for complete academic control

Student Features

- Attendance tracking
- Live timetable
- Assignment submission
- View marks and performance
- Notifications and alerts
- Email updates

Complete academic visibility for students

Faculty Features

- Digital attendance marking
- Broadcast announcements
- Low attendance alerts
- Assignment management
- Upload and manage internal marks

Better control + full transparency

Comparison with Existing Systems

Feature	WhatsApp	Google Classroom	ClassHub
Organized Updates	X	O	O
Attendance	X	X	O
Timetable	X	X	O
Assignments	X	O	O
Internal Marks	X	X	O

ClassHub = Complete academic system

Design Inspiration: StudyMate

Inspired by structured platforms like StudyMate

- Role-based dashboards
- Clean, intuitive UI
- Feature-specific modules
- Problem-focused design

Leveraging proven platform patterns

Technologies Used

Frontend

- HTML5
- CSS3
- JavaScript

Backend

- Node.js
- Express.js

Database

- MongoDB

Security

- JWT, Bcrypt
- Nodemailer, Multer

Modern stack for scalability and reliability

Project Status and Deployment

Fully Completed

- All features implemented
- Hosted on AWS
- Real-time working system

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Conclusion

- Eliminates WhatsApp chaos
- Centralizes academic management
- Improves transparency
- Secure and scalable
- User-friendly interface

One Platform for Complete Academic